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Spin-Torsion Bounce Cosmology: Fourteen Structural Barriers and One Surviving Prediction

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The Einstein-Cartan-Holst (ECH) framework produces a nonsingular quantum bounce at critical density $\rho_{\rm crit} \approx 0.27\,\rho_{\rm Pl}$ and generates a parity-odd four-fermion interaction through the Barbero-Immirzi parameter $\gamma = 0.274$. We systematically assess this framework's cosmological predictions across fifteen logical routes (Branches A–O) spanning three domains: direct dark energy derivation, observable signature generation, and vacuum state selection.

All routes encounter structural barriers—mechanism-independent obstructions rooted in symmetry, scaling, or conservation law. We catalog fourteen such barriers organized into five failure modes: (I) geometric specificity lost on FRW backgrounds, (II) 10^{122} scale separation between bounce and dark energy, (III) generic or undetectable signatures, (IV) Hamiltonian phase-space conservation, and (V) gravitational democracy at the Planck scale. No first-principles route from the bounce to dark energy survives, and no distinctive bounce signature is accessible to current or planned experiments.

One indirect testable prediction remains. The framework's Planck-scale parity-odd sector motivates an effective spectator axion-like particle with $f_a \sim M_{\rm Pl}$ and Standard Model anomaly coupling $C_{a\gamma} = 8$. This ALP predicts cosmic birefringence $\beta = C_{a\gamma}\alpha_{\rm em}\,\theta_i/(4\pi) \approx 0.27^\circ$ for $\mathcal{O}(1)$ initial misalignment, consistent with the observed $0.342 \pm 0.094^\circ$ (Planck PR4 + ACT DR6; 3.6σ vs. $\beta = 0$). MCMC analysis yields $\theta_i = 1.36 \pm 0.44$ with no fine-tuning. The ALP model is statistically equivalent to a free birefringence parameter ($\Delta\mathrm{AIC} = +2$) but provides physical interpretation: natural $\mathcal{O}(1)$ parameters, f_a-independent prediction, and falsifiable structure. We note that spectator ALP models with similar conclusions have been studied previously; our contribution is the identification of this prediction as the sole surviving testable output of a comprehensive framework assessment. LiteBIRD ($\sigma_\beta \approx 0.01^\circ$) will be decisive. Dark energy is treated as a cosmological constant Λ; the ALP contributes negligible energy density.

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CONTENTS

I. INTRODUCTION

The cosmological constant problem—the $\sim 10^{122}$ discrepancy between the observed vacuum energy density and quantum field theory expectations [1]—remains

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among the deepest puzzles in fundamental physics. The Einstein-Cartan-Holst (ECH) framework, which extends general relativity by coupling spacetime torsion to fermion spin, offers a geometrically rich arena in which to search for new cosmological phenomena [2, 3]. Combined with loop quantum gravity corrections, ECH produces a nonsingular quantum bounce that replaces the classical Big Bang singularity [4–6].

Recent observations of cosmic birefringence—a uniform rotation of CMB polarization angles—provide tantalizing evidence for parity violation in the gravitational or dark sector. The combined Planck PR4 and ACT DR6 analysis yields $\beta = 0.342 \pm 0.094°$, excluding $\beta = 0$ at 3.6σ [7–9].

This paper presents a comprehensive assessment of the ECH framework's cosmological predictions. Our approach is systematic: we identify all logical routes from the bounce to late-time observables, pursue each until it produces a testable prediction or encounters a structural barrier, and catalog the results. The outcome is summarized in Table I.

TABLE I. Executive summary.

Domain	Result	Status
Nonsingular bounce	$\rho_{\rm crit} \approx 0.27\,\rho_{\rm Pl}$	Viable
DE from bounce	14 barriers	Closed
Distinctive signatures	Generic or undetectable	Closed
ALP birefringence	$\beta \approx 0.27°$	Consistent

A. Original contributions

1. Complete assessment of ECH bounce cosmology across fifteen branches (A–O), yielding fourteen structural barriers organized into five failure modes.

2. Systematic closure of all first-principles routes from the bounce to dark energy and distinctive observational signatures.

3. Identification of spectator ALP birefringence as the sole surviving testable prediction, with MCMC constraints on (θ_i, m_a).

4. Explicit demonstration that the ALP model is statistically equivalent to a free β parameter—physical interpretation rather than statistical improvement.

5. Falsification program: LiteBIRD will determine θ_i to ± 0.04 rad, providing a decisive test.

B. Paper organization

Section II reviews the ECH framework and bounce. Section III presents the structural closure assessment. Section IV derives the ALP birefringence prediction and MCMC constraints. Section V discusses experimental prospects. Section VI provides context and limitations. Section VII concludes.

II. THEORETICAL FRAMEWORK

A. The Einstein-Cartan-Holst action

The ECH action generalizes the Einstein-Hilbert action by incorporating torsion and the Holst (topological) term [10]:

$$S_{\rm ECH} = \frac{1}{16\pi G}\int \left(e^a \wedge e^b \wedge R_{ab} + \frac{1}{\gamma}\, e^a \wedge e^b \wedge {}^*R_{ab}\right) + S_{\rm matter}, \tag{1}$$

where γ is the Barbero-Immirzi parameter, fixed by loop quantum gravity black hole entropy calculations to $\gamma = 0.274$ [11–13].

Integrating out torsion algebraically yields a parity-odd four-fermion interaction [2, 3]:

$$\mathcal{L}_{\rm torsion} = -\frac{3\pi G}{2}\,\frac{\gamma^2}{1+\gamma^2}\,(J_5^\mu)(J_{5\mu}), \tag{2}$$

where $J_5^\mu = \bar{\psi}\gamma^\mu\gamma_5\psi$ is the axial fermion current.

B. Nonsingular quantum bounce

Loop quantum gravity corrections modify the Friedmann equation at Planck-scale densities [4, 5]:

$$H^2 = \frac{8\pi G}{3}\,\rho\left(1 - \frac{\rho}{\rho_{\rm crit}}\right), \tag{3}$$

with critical density $\rho_{\rm crit} \approx 0.27\,\rho_{\rm Pl}$. At $\rho = \rho_{\rm crit}$, $H = 0$ and the universe undergoes a nonsingular bounce. The scale factor evolves as $a(t) \propto (1 + 4\alpha^2 t^2)^{1/4}$ near the bounce, where $\alpha \equiv \sqrt{8\pi G\rho_{\rm crit}/3}$.

The bounce is time-reversal symmetric: contracting and expanding phases are mirror images. This symmetry has important consequences for perturbation transfer (Sec. III).

C. Parity-odd sector and late-time phenomenology

The $(J_5)^2$ interaction (2) generates, at one loop, a finite parity-violating coefficient of natural order $\mathcal{O}(\alpha_{\rm em}/M_{\rm Pl})$ [14, 15]. Whether this coefficient persists as a late-time vacuum term is an open question. Four explicit routes to deriving $w = -1$ from the minimal model have been tested and closed (Branches A–D in Sec. III).

We therefore treat dark energy as a separate cosmological constant Λ. The parity-odd structure motivates an effective axion-like particle sector, developed in Sec. IV.

III. STRUCTURAL CLOSURE OF THE MINIMAL FRAMEWORK

We systematically investigate all logical routes from the ECH bounce to late-time observables, organizing them into fifteen branches (A–O). Each branch is pursued until it produces a distinctive, testable prediction or encounters a structural barrier—a mechanism-independent obstruction rooted in symmetry, scaling, or conservation law.

A. Methodology

Candidate connections are classified into three tiers: *direct derivation* of dark energy (Branches A–G), *observable signatures* (Branches H, K–M), and *state selection and vacuum determination* (Branches I, J, N, O). A branch is closed when a structural barrier is identified; it is classified as *generic* when the result is shared by all bouncing cosmologies.

B. Fourteen structural barriers

Figure 1 catalogs all fourteen barriers. We organize them into five failure modes.

Mode I: Geometric specificity lost on FRW (Barriers 1–4, 11). The ECH Lagrangian contains rich geometric structure—propagating torsion, the Nieh-Yan topological density, non-metricity, disformal couplings. On a Friedmann-Robertson-Walker background, however, isotropy and homogeneity set $T_\mu = Q_\mu = 0$. The only surviving geometric scalar is $R(g)$, identical in GR and all metric-compatible theories. Specifically: the PGT torsion mode's mass and coupling are locked by a single parameter (Barrier 1); shift symmetry and geometric content are mutually exclusive for the Nieh-Yan pseudoscalar (Barrier 2); any torsion-descended scalar reduces to a standard scalar-tensor field on FRW (Barrier 3); disformal couplings require two field gradients but the connection provides only one, giving dimension-8 suppression $\sim 10^{-122}$ (Barrier 4).

Mode II: Scale separation (Barriers 5–7). The bounce occurs at $\rho_{\rm crit} \sim 0.27\,\rho_{\rm Pl}$, while dark energy operates at $\rho_{\rm DE} \sim 10^{-122}\,\rho_{\rm Pl}$. This 10^{122} hierarchy prevents communication: the bounce contributes $\sim 10^{-32}$ of spacetime volume integrals (Barrier 5); quintessence attractors erase all bounce information or require 10^{-30} precision that the bounce cannot provide (Barrier 6); action parameters cannot be constrained by boundary conditions, and cyclic cosmology with $\Lambda > 0$ is geometrically incompatible (Barrier 7).

Mode III: Generic or undetectable signatures (Barriers 8, 10, 12). The four-fermion interaction $(J_5)^2$ is parity-*even* (pseudovector squared = scalar), and $\tilde{R}R = 0$ on FRW: no tensor chirality signal exists (Barrier 8). The tensor power spectrum $P_T \sim 2 \times 10^{-64}$ and scalar transfer function $\mathcal{T}(k) = 1$ for $k \ll k_b$ are calculable but either undetectable or shared by all symmetric bounces. Generic UV→IR mechanisms operate in any hot cosmology (Barrier 10), and the vacuum amplification ceiling $\Omega_{\rm GW} \propto (H_b/M_{\rm Pl})^2$ applies universally (Barrier 12).

Mode IV: Hamiltonian conservation (Barrier 9). The bounce is a conservative scattering event at $H = 0$. Liouville's theorem preserves phase-space volume: the bounce *rotates* dark-sector states but cannot *select* them.

Mode V: Gravitational democracy (Barriers 13–14). At $T \sim M_{\rm Pl}$, torsion is one of ~ 100 gravitational-strength channels ($\sim 1\%$ contribution), conserving B and L. Irreversible processes can be triggered by the bounce, but the resulting vacuum energy is set by hidden-sector parameters decoupled from the trigger: the bounce controls *whether* a transition occurs, not *what* vacuum energy results.

C. Branch status summary

TABLE II. Status of all fifteen branches.

Branch	Route tested	Barriers	Status
A	Propagating torsion (PGT)	1	Closed
B	Nieh-Yan pseudoscalar	2	Closed
C	Environmental mass	3	Closed
D	Disformal coupling	4	Closed
E	Global vacuum integrals	5	Closed
F	Bounce-set initial conditions	6	Closed
G	Cyclic vacuum selection	7	Closed
H	Tensor spectrum & chirality	8	Closed
I	Horndeski DE stability	—	Weak
J	State selection	9	Closed
K	Scalar perturbations	—	Generic
L	UV-IR bridge (PGT)	10, 11	Closed
M	PGT bounce GW spectrum	12	Generic
N	Baryogenesis & relics	13	Closed
O	Hidden-sector vacuum	14	Closed

D. Implications

The fourteen barriers establish that the minimal ECH framework is *observationally inert* in the direct sector. The bounce is trapped between three regimes: too high-energy for late-time observables (Modes I, II), too generic at its own scale (Modes III, V), and too decoupled from late-time vacuum (Modes IV, V).

These results do not invalidate ECH. The bounce remains a well-defined, singularity-free cosmology. What they establish is that dark energy requires separate input,

Fourteen Structural Barriers of ECH Bounce Cosmology

#	Barrier	Branch	Physical mechanism	Failure mode
1	Mass-coupling lock	A	*m and g locked by single PGT parameter t_3*	I
2	Topological-shift duality	B	*Shift symmetry ↔ geometric content exclusive*	I
3	Scalar-tensor universality	C	*$T_0 = Q_0 = 0$ on FRW; only R(g) survives*	I
4	Planck suppression	D	*1 $\partial\varphi$/vertex; disformal needs 2; dim-8 ~ 10^{-122}*	I
5	Scale separation	E	*Bounce ~ 10^{-32} of spacetime volume integrals*	II
6	Attractor-sensitivity	F	*Attractors erase φ_i; otherwise 10^{-30} precision needed*	II
7	Parameter immunity	G	*Action params ≠ boundary conditions; cyclic incompatible*	II
8	Parity-even interaction	H	*$(J^5)^2$ is parity-even; $\tilde{R}R = 0$ on FRW*	III
9	Phase-space conservation	J	*Liouville theorem at H = 0; no contraction*	IV
10	UV–IR specificity	L	*Generic mechanisms don't need the bounce*	III
11	Decoupling universality	L	*Light torsion: $g_{eff} \sim m_T / M^2_{Pl}$*	I
12	Vacuum amplification ceiling	M	*$\Omega_{GW} \propto (H_b/M_{Pl})^2$; gap ≥ 10^{17} to detectors*	III
13	Gravitational democracy	N	*Torsion ~ 1% of Planck-era channels; B, L conserved*	V
14	Bounce–vacuum decoupling	O	*Trigger (bounce) ≠ outcome (ρ_{DE})*	V

Failure modes: I: Geometric specificity lost on FRW; II: Scale separation (10^{122} hierarchy); III: Generic or undetectable; IV: Hamiltonian conservation; V: Gravitational democracy

FIG. 1. The fourteen structural barriers identified across fifteen branches of ECH cosmology. Color-coded badges indicate the failure mode classification (I–V). Each barrier represents a mechanism-independent obstruction that cannot be removed by parameter adjustment within the minimal framework.

and no distinctive bounce signature is accessible. One indirect handle survives: the parity-odd sector motivates a Planck-scale ALP whose photon coupling produces cosmic birefringence.

IV. SPECTATOR ALP BIREFRINGENCE: THE SURVIVING PREDICTION

The structural closure eliminates all direct routes from the bounce to late-time observables. One indirect testable handle remains: cosmic birefringence via a Planck-scale ALP motivated by the ECH parity-odd sector. We stress that this ALP is not uniquely derived from ECH—any Planck-scale pseudoscalar with Standard Model anomaly coupling yields the same prediction [16, 17]. ECH provides one possible UV completion.

A. The effective spectator ALP

The one-loop parity-violating coefficient of the ECH sector [14, 15] motivates an effective pseudoscalar:

$$\mathcal{L}_{\rm ALP} = -\tfrac{1}{2}(\partial\phi)^2 - m_a^2 f_a^2\left[1-\cos\frac{\phi}{f_a}\right] - \frac{g_{a\gamma}}{4}\,\phi F\tilde{F}, \quad (4)$$

with $g_{a\gamma} = C_{a\gamma}\alpha_{\rm em}/(2\pi f_a)$. We fix $f_a = M_{\rm Pl} = 2.435\times 10^{18}$ GeV and $C_{a\gamma} = 8$ (SM fermion content), leaving a two-parameter model: θ_i (initial misalignment) and m_a (ALP mass).

The spectator regime ($m_a >$ few $\times\, H_0$) ensures the ALP has oscillated away: $\Omega_a \ll 0.01$ today, with no effect on background expansion. This avoids the rolling-versus-freezing tension [18]: a single field cannot simultaneously provide $w \approx -1$ dark energy and large birefringence; the maximum $\beta \approx 0.16^\circ$ on the $\Omega_a = 0.68$ contour is a factor of two below observation.

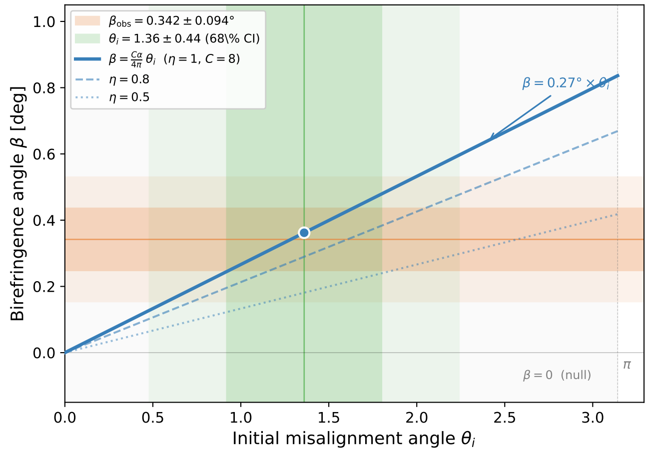

FIG. 2. Birefringence angle β versus initial misalignment θ_i for the spectator ALP with $C = 8$ and $\eta = 1$ (solid), 0.8 (dashed), and 0.5 (dotted). The orange band shows the observation $\beta_{\rm obs} = 0.342 \pm 0.094°$; the green band shows the MCMC posterior $\theta_i = 1.36 \pm 0.44$.

B. Birefringence prediction

As the ALP evolves from θ_i at recombination to θ_0 today, CMB photons accumulate polarization rotation:

$$\beta = \frac{C_{a\gamma}\alpha_{\rm em}\,\theta_i\,\eta}{4\pi}, \tag{5}$$

where $\eta \equiv [\theta(z_{\rm rec}) - \theta(0)]/\theta_i$ is the rolling efficiency, computed numerically via DOP853 integration [19] from $z_{\rm rec} = 1089.8$ to $z = 0$.

Crucially, f_a cancels exactly: β depends only on $C_{a\gamma}$, $\alpha_{\rm em}$, θ_i, and $\eta(m_a/H_0)$. For $C = 8$, $\theta_i = 1$, $\eta \to 1$:

$$\beta \approx 0.27°, \tag{6}$$

in good agreement with $\beta_{\rm obs} = 0.342 \pm 0.094°$ [7, 8]. Figure 2 shows the prediction as a function of θ_i.

C. MCMC constraints

We constrain $(\theta_i, \log_{10}(m_a/{\rm eV}))$ using a Gaussian likelihood on $\beta_{\rm obs} = 0.342 \pm 0.094°$, sampled with Cobaya [20] (4 chains, $\hat{R} - 1 < 0.01$, flat priors $\theta_i \in [0.01, \pi]$, $\log_{10} m \in [-33, -28]$).

We note this uses the summary statistic $\beta_{\rm obs} \pm \sigma_\beta$ rather than the full C_ℓ^{EB} spectrum employed in Refs. [21, 22]. For the spectator regime ($\eta \to 1$), birefringence is ℓ-independent and the summary statistic captures the full information content.

After 2160 accepted samples ($\hat{R} - 1 = 0.008$, $N_{\rm eff} > 1000$):
The inferred $\theta_i = 1.36 \pm 0.44$ is $\mathcal{O}(1)$, consistent with random initial conditions on $[0, 2\pi)$. No fine-tuning is required. The mass is weakly constrained from below ($m_a \gtrsim 3\,H_0$) but unconstrained from above.

TABLE III. Posterior for the spectator ALP ($f_a = M_{\rm Pl}$, $C = 8$).

Parameter	Mean ± 68%	95% CI
θ_i	1.36 ± 0.44	[0.47, 2.28]
$\log_{10}(m_a/{\rm eV})$	-31.3 ± 0.7	[−32.4, −30.0]
β [deg]	0.336 ± 0.107	[0.13, 0.52]
η	0.957 (median 0.999)	[0.40, 1.17]

D. Model comparison

We compare three models: (0) free β (1 parameter), (2) ALP with $C = 8$ (2 parameters), (2b) ALP with C free (3 parameters).

TABLE IV. Model comparison. All non-null models achieve $\chi^2_{\rm min} \approx 0$.

Model	k	$\Delta\chi^2$ vs null	ΔAIC vs free-β
Null ($\beta = 0$)	0	—	—
Free β	1	13.2 (3.6σ)	0.0
ALP (C=8)	2	13.2 (3.6σ)	+2.0
ALP (C free)	3	13.2 (3.6σ)	+4.0

All three produce identical β posteriors and identical 3.6σ detection significance. The ALP does not improve the fit. Floating $C_{a\gamma}$ reveals a degeneracy: $C \times \theta_i \approx 10.6$ (correlation $r = -0.52$) with the SM value $C = 8$ comfortably within the posterior.

E. Assessment: physical interpretation, not statistical improvement

The ALP provides no statistical improvement over free β—expected with one Gaussian data point. Its value is physical:

1. *Natural parameters.* $\theta_i \approx 1.3$ is $\mathcal{O}(1)$; free $\beta = 0.34°$ has no such interpretation.

2. *Structural prediction.* β is determined by C, α, θ_i, and η—not a free phenomenological parameter.

3. *Falsifiability.* LiteBIRD [23] with $\sigma_\beta \approx 0.01°$ will determine θ_i to ± 0.04 rad. The model is also falsified by frequency-dependent or anisotropic birefringence.

4. *Framework context.* Unlike previous ALP birefringence studies [16, 17, 24], this prediction emerges as the sole survivor of a fifteen-branch assessment.

Spectator ALP models with similar conclusions have been studied by Fujita *et al.* [16] and Nakagawa *et al.* [17]. Our contribution is the ECH-motivated two-parameter reduction, the explicit free-β baseline comparison, and the identification as the sole surviving prediction.

V. EXPERIMENTAL CONSTRAINTS AND FORECASTS

A. Existing bounds

All laboratory, astrophysical, and cosmological bounds are satisfied with margins of 9–32 orders of magnitude. The model coupling $g_{a\gamma} = 3.8 \times 10^{-21}$ GeV^{-1} lies far below CAST [25] ($g < 6.6 \times 10^{-11}$ GeV^{-1}) and SN1987A ($g < 5 \times 10^{-12}$ GeV^{-1}). The model mass $m_a \sim 10^{-31}$ eV is below the black hole superradiance exclusion window and below spectral distortion sensitivity.

The isocurvature constraint ($f_a < 10^{19}$ GeV for $m \sim H_0$) is marginally satisfied at $f_a = M_{\rm Pl}$, and the dark matter overproduction bound ($\theta_i < 2.4$ for $f_a = M_{\rm Pl}$) is satisfied by $\theta_i = 1.3$.

B. LiteBIRD forecast

LiteBIRD [23], with projected $\sigma_\beta \approx 0.01°$, provides a decisive test:

- If $\beta = 0.27°$: detected at $> 20\sigma$.
- If $\beta = 0$: model excluded at $> 20\sigma$.
- θ_i determined to ± 0.04 rad.

C. Additional tests

Ground-based CMB-S4 measurements of anisotropic birefringence and frequency dependence can distinguish ALP rotation from Faraday rotation and instrumental systematics [26]. DESI DR2 constraints on $w(z)$ will test whether a subdominant axion contributes to dark energy dynamics [17, 27].

VI. DISCUSSION

A. The closure in context

The fourteen barriers span the full space of logical connections between the ECH bounce and late-time physics. The obstruction is not a single no-go theorem but a *web* of independent barriers, each rooted in different physics (symmetry, scaling, conservation, universality). Overcoming any one barrier does not help, because adjacent barriers remain.

This comprehensive closure is, to our knowledge, the first such systematic assessment for any specific quantum gravity cosmology. While individual no-go results exist in the literature (e.g., for loop quantum cosmology perturbation transfer), no previous work has mapped the entire space of possible predictions for a given framework.

B. What the ALP provides—and what it does not

The spectator ALP with $f_a = M_{\rm Pl}$ and $C = 8$ provides a clean physical interpretation of the observed birefringence: the rotation angle $\beta \approx 0.27°$ arises from an $\mathcal{O}(1)$ initial condition with no free parameters beyond θ_i and m_a.

However, the ALP is not uniquely derived from ECH. Any Planck-scale pseudoscalar—arising from string compactifications, Peccei-Quinn symmetry breaking at $M_{\rm Pl}$, or the torsion-descended axion [28]—yields the same prediction. ECH motivates $f_a \sim M_{\rm Pl}$ through its parity-odd sector, but this motivation is not unique.

The ALP also does not explain dark energy. In the spectator regime, the field's energy density is negligible. Λ is assumed, not derived. The mass $m_a \sim$ few $\times\, H_0$ is a free parameter subject to the cosmological coincidence problem.

C. Limitations

1. $w = -1$ is assumed (Λ), not derived from ECH.
2. The ALP-photon coupling is motivated but not derived from a first-principles one-loop calculation in ECH.
3. The one effective data point ($\beta_{\rm obs}$) constrains one combination of parameters; the ALP is statistically equivalent to free β.
4. The mass m_a is weakly constrained by our analysis; the full C_ℓ^{EB} spectrum treatment of Ref. [22] provides superior mass discrimination.

VII. CONCLUSIONS

The Einstein-Cartan-Holst framework produces a well-defined nonsingular quantum bounce at $\rho_{\rm crit} \approx 0.27\, \rho_{\rm Pl}$. Systematic investigation of fifteen routes from this bounce to late-time observables identifies fourteen structural barriers spanning five failure modes. No first-principles derivation of dark energy survives, and no distinctive bounce signature is accessible to current or planned experiments. The bounce is theoretically viable but observationally inert in the direct sector.

One indirect prediction survives. A spectator ALP motivated by the framework's Planck-scale parity-odd sector, with $f_a = M_{\rm Pl}$ and SM anomaly coupling $C = 8$, predicts $\beta \approx 0.27°$ for $\mathcal{O}(1)$ initial misalignment. MCMC analysis yields $\theta_i = 1.36 \pm 0.44$, consistent with observation ($\beta_{\rm obs} = 0.342 \pm 0.094°$; 3.6σ). The model is statistically equivalent to a free β parameter but provides physical interpretation: natural parameters, f_a-independent prediction, and falsifiable structure.

LiteBIRD will be decisive. With $\sigma_\beta \approx 0.01^\circ$, the predicted $\beta \approx 0.27^\circ$ is detected at $> 20\sigma$ or excluded at comparable significance. Dark energy remains unexplained by the minimal ECH framework. The ALP birefringence prediction is the single testable output of this program.

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